APMPX4 CAN pinout

Every connector, header and solder pad on the STM32L433 CAN board, read from the EasyEDA PCB and netlist, with what firmware 0.5.39 does with each pin. It comes in two builds: `apmpx4_can_l433rc_dronecan` is the GNSS filter, and `apmpx4_can_l433rc_dronecan_nognss` is the same firmware without GNSS support, which leaves CN2 unused.

BOARD	MCU	CAN	POWER IN	CONNECTORS	PROGRAMMING
39.3 × 29.2 mm	**STM32L433RCT6, 80 MHz**	**TJA1051 · 1 Mbit/s · node 42**	**5 V on any connector pin 1**	**JST GH 1.25 mm, Pixhawk order**	**SWD on H1, no USB**

Top side · connectors and wiring

The board is drawn to scale from the PCB placement, wired to a Pixhawk over DroneCAN and to a u-blox ZED-F9P over UART. The dark lip on each GH connector marks the cable opening; pin 1 is the 5 V pin on every GH connector and the square pad on every 2.54 mm header.

u-blox ZED-F9P

UART1, Pixhawk-style GH6 cable, pin 1 to pin 1

```
TX2 · PA2   →  F9P UART1 RX
RX2 · PA3   ←  F9P UART1 TX
autoconfigured to 460800 baud
compass on SCL/SDA reaches I²C3,
which 0.5.39 does not read
```

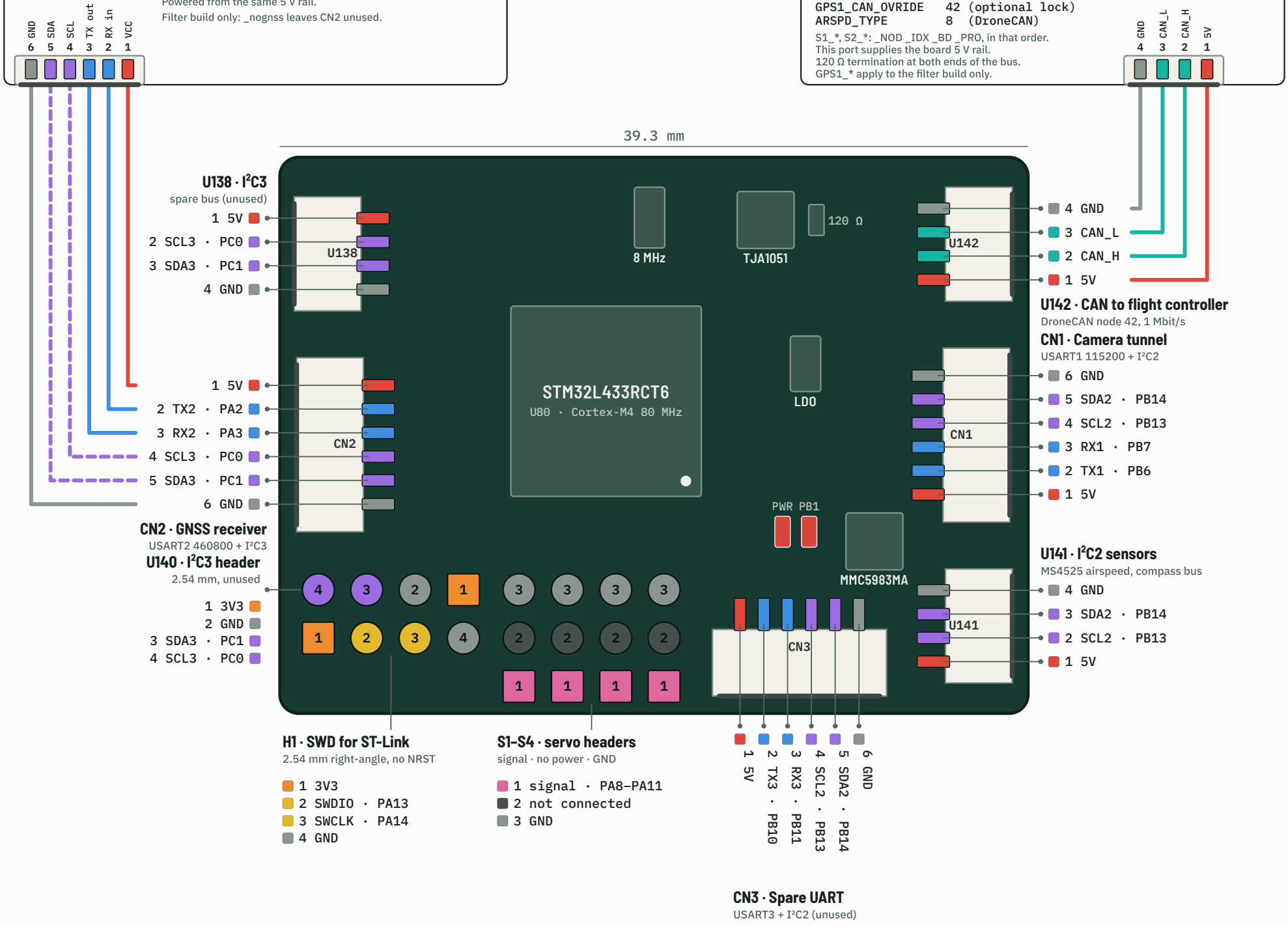
Powered from the same 5 V rail.
Filter build only: _nognss leaves CN2 unused.

Pixhawk · ArduPilot

CAN1 port, straight GH4 cable · DroneCAN 1 Mbit/s

```
CAN_P1_DRIVER      1
CAN_P1_BITRATE     1000000
CAN_D1_PROTOCOL    1  (DroneCAN)
CAN_D1_UC_SER_EN   1
CAN_D1_UC_S1_*     42 0 115 2  (camera)
CAN_D1_UC_S2_*     42 1 115 2  (filter)
MAV_SYSID          1  (4.6: SYSID_THISMAV)
GPS1_TYPE          9  (DroneCAN)
GPS1_CAN_OVRIDE    42 (optional lock)
ARSPD_TYPE         8  (DroneCAN)
```

S1_*, S2_*: _NOD _IDX _BD _PRO, in that order.
This port supplies the board 5 V rail.
120 Ω termination at both ends of the bus.
GPS1_* apply to the filter build only.

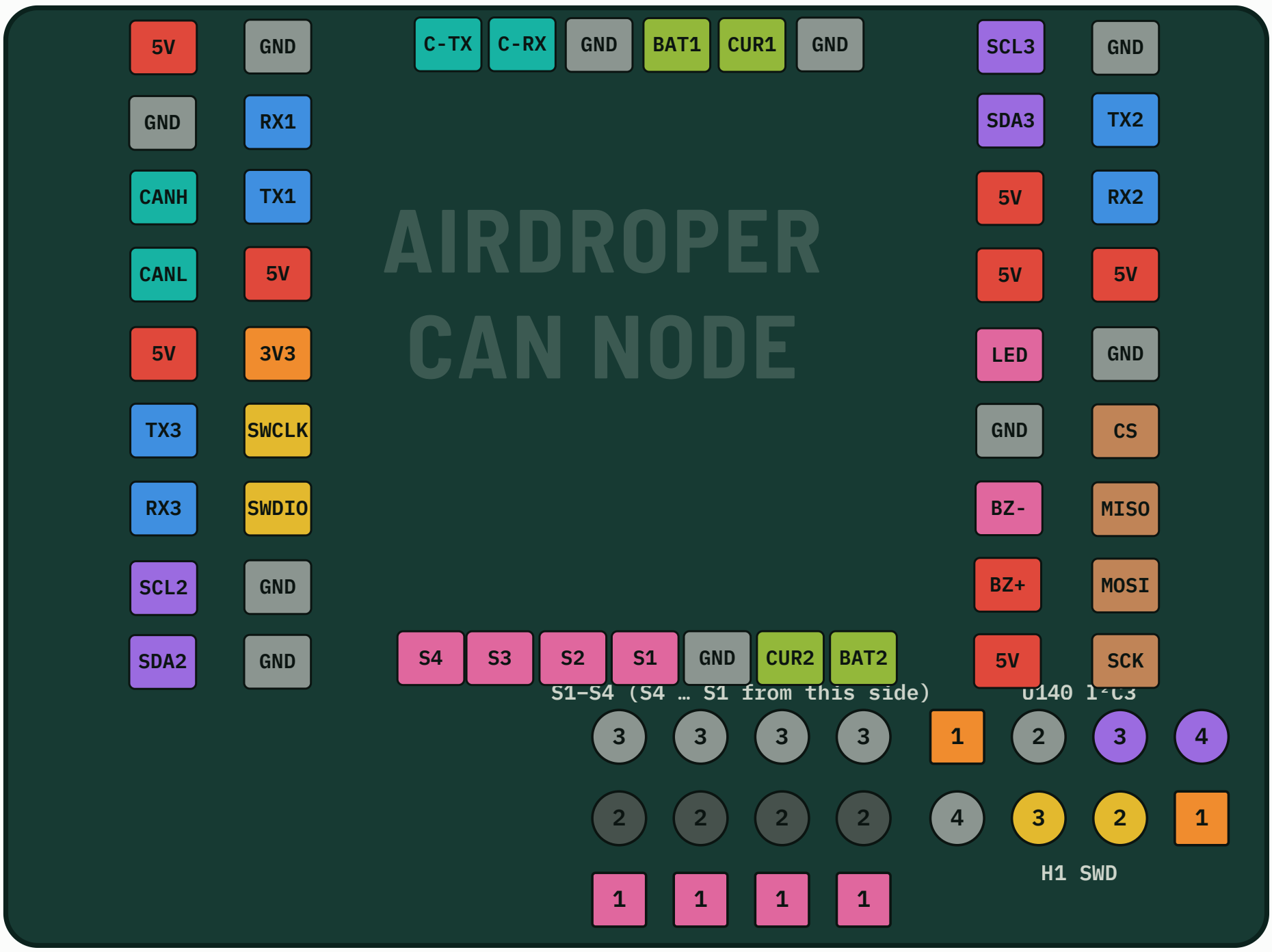


5 V rail · 3.3 V · Ground · CAN · UART · I²C · SWD debug · Analog in · PWM / GPIO · SPI · Not connected

Top view with the flight controller and the GNSS receiver; the airspeed sensor and the camera are in the hook-up cards below. Both cables are straight Pixhawk-style cables: CN2 has the same pin order as a Pixhawk GPS port and U142 as a Pixhawk CAN port, so the TX/RX cross happens inside the pin assignment, not the cable. Dashed wires are connected but not read by firmware 0.5.39. Labels read *pin · signal · STM32 pin*.

Bottom side · solder pads

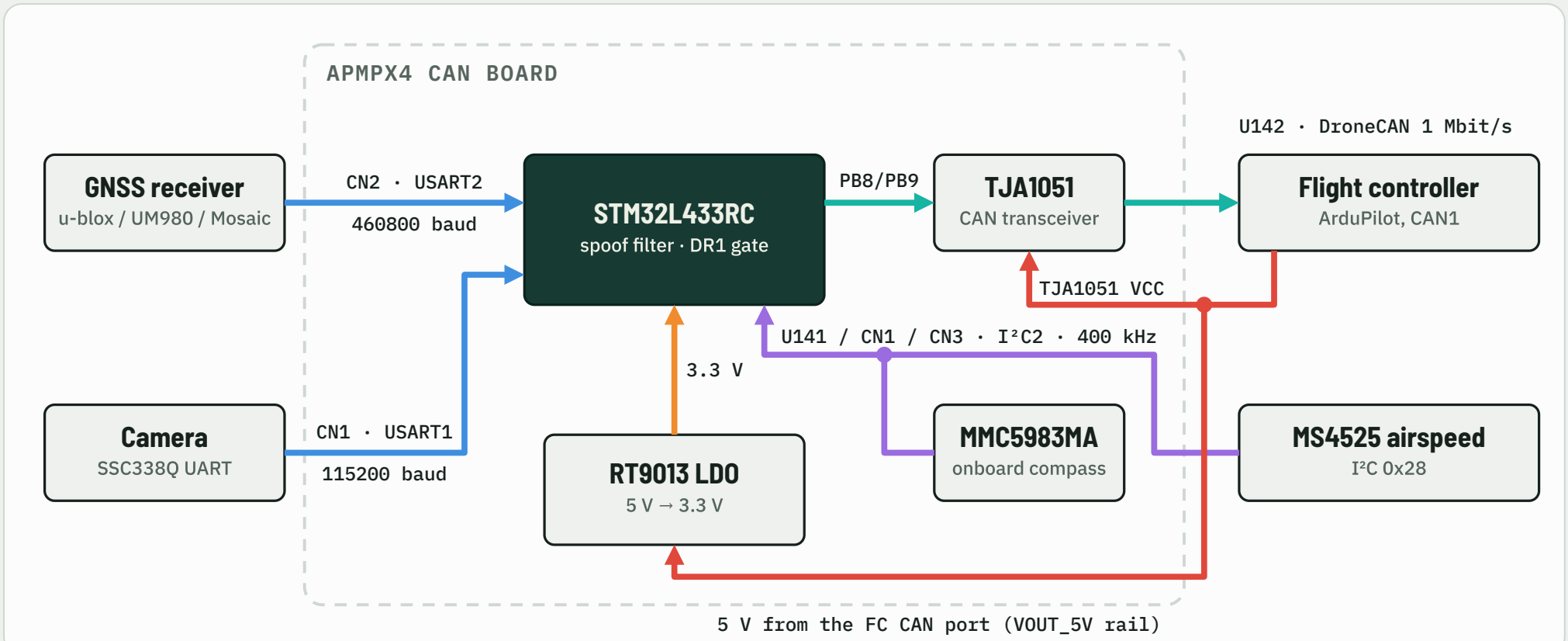
The 49 pads on the bottom carry the same nets as the connectors plus the extras (SPI, battery sense, buzzer, LED, servo outputs). Shown the way you hold the board to solder: turned over left to right, so the header block sits at the bottom right. Pads are enlarged for legibility; names match the silkscreen.



Bottom view, mirrored. Real pads are 1.5 mm squares on a 2.4 mm pitch.

How it is wired

GNSS data comes in on CN2, the filter decides whether it may pass, and it leaves as DroneCAN on U142. The flight controller's CAN port also feeds the 5 V rail that the transceiver and the 3.3 V regulator run from.

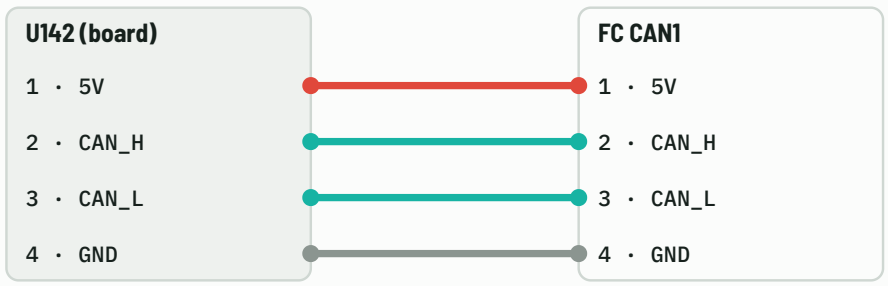


Signal and power flow for firmware 0.5.39. Arrows point in the direction data or power moves. The _nognss build has no GNSS path; CN2 is unused.

Hook-up for bench tests

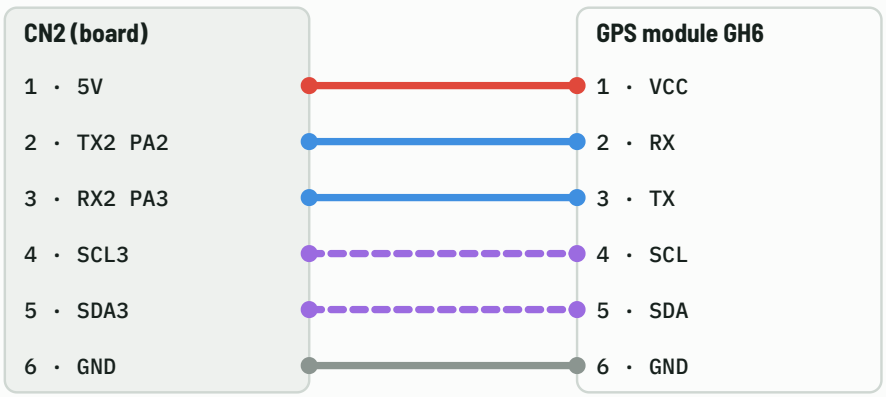
Dashed wires are optional or not read by the firmware.

Flight controller CAN



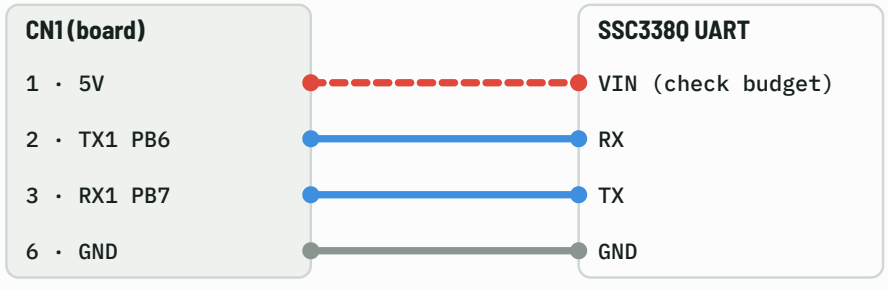
Straight Pixhawk CAN cable, pin 1 to pin 1. This link also powers the board.

GNSS receiver



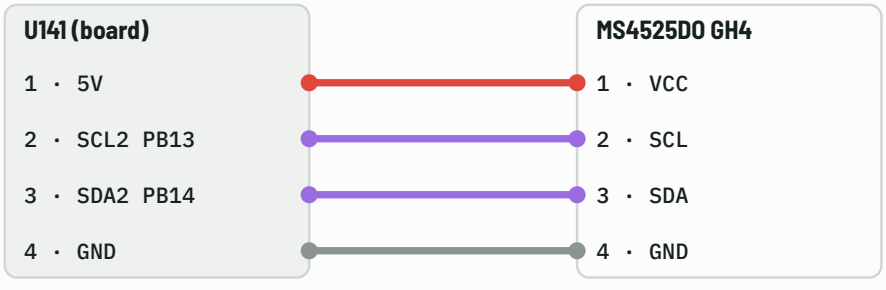
Straight Pixhawk GPS cable. Board TX2 lands on the module RX, RX2 on its TX. The module compass on pins 4–5 reaches I²C3, which firmware 0.5.39 does not read. Filter build only.

Camera (MAVLink tunnel)



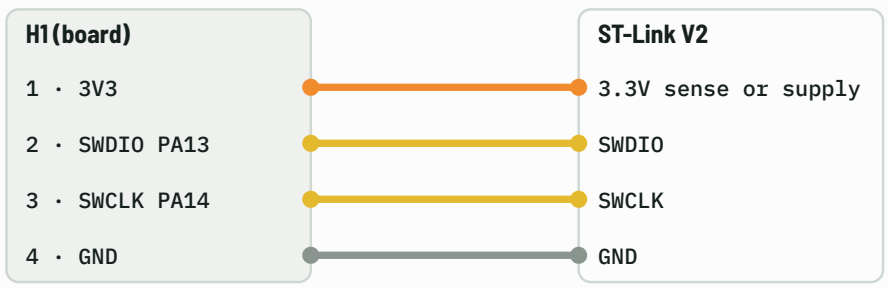
Cross TX/RX by function. Power the camera from its own supply unless its draw fits the FC CAN port budget; leave CN1 pins 4–5 free, they are the sensor bus.

Airspeed sensor



A 5 V Pixhawk-style module with its own 3.3 V regulator, straight cable. The bus is 3.3 V and shared with the onboard MMC5983MA at 0x30.

ST-Link V2 (SWD)



Connect in Normal or Hot-plug mode: H1 has no NRST. Pin 1: a genuine ST-LINK senses the target voltage there, so always connect it; a clone supplies 3.3 V there, so connect it only when nothing else powers the board.

Before you power it

One 5 V rail

Pin 1 of every GH connector and every **5V** pad is the same `VOUT_5V` net. Feed it from exactly one source, normally the flight controller's CAN port.

ST-Link power is not enough for CAN

The TJA1051 runs from the 5 V rail. Powered only from the ST-Link's 3.3 V, the MCU runs but CAN goes bus-off.

Termination always on

R56 (120 Ω) is fitted permanently. Place the board at one end of the CAN bus.

Schematic note vs firmware

The schematic's note reads "UART1 debug, UART2 MSP, UART3 GPS". Firmware 0.5.39 instead takes GNSS on **CN2 / USART2** and the camera on **CN1 / USART1**; CN3 / USART3 is spare.

Servo headers carry no power

S1–S4 are signal, not connected, GND. Supply servo power separately.

Battery inputs

BAT1 and BAT2 divide by 21 (about 69 V full scale) with the 20 kΩ the schematic shows, or by 11 (about 36 V) with the 10 kΩ part it lists: check R170 / R173. CUR1 and CUR2 go straight to the ADC: 0–3.3 V only. Not read by 0.5.39.

Two firmware builds

`apmpx4_can_l433rc_dronecan` is the GNSS filter. `apmpx4_can_l433rc_dronecan_nognss` is the same firmware without GNSS support: airspeed, compass, camera tunnel, fuel and parameters on node 42, with CN2 unused.

I²C sensors are optional

The MS4525 and the compass are both in the firmware. A sensor that does not answer after power-up counts as not fitted and raises no fault. One that answered and then stops does, and ArduPilot reports "Node 42 unhealthy!".

Pin tables

Connectors first, then the bottom pads, then the MCU pins by function.

U142 · CAN to flight controller

used by 0.5.39

DroneCAN node 42, 1 Mbit/s · JST GH 1.25 mm

PIN	SIGNAL	NET	MCU
1	5V	VOUT_5V	
2	CAN_H	CAN1_H	TJA1051
3	CAN_L	CAN1_L	TJA1051
4	GND	GND	

CN2 · GNSS receiver

filter build only

USART2 460800 + I²C3 · JST GH 1.25 mm

PIN	SIGNAL	NET	MCU
1	5V	VOUT_5V	
2	TX2	USART2_TX	PA2
3	RX2	USART2_RX	PA3
4	SCL3	I2C3_SCL	PC0
5	SDA3	I2C3_SDA	PC1
6	GND	GND	

CN1 · Camera tunnel

used by 0.5.39

USART1 115200 + I²C2 · JST GH 1.25 mm

PIN	SIGNAL	NET	MCU
1	5V	VOUT_5V	
2	TX1	USART1_TX	PB6
3	RX1	USART1_RX	PB7
4	SCL2	I2C2_SCL	PB13
5	SDA2	I2C2_SDA	PB14
6	GND	GND	

U141 · I²C2 sensors

used by 0.5.39

MS4525 airspeed, compass bus · JST GH 1.25 mm

PIN	SIGNAL	NET	MCU
1	5V	VOUT_5V	
2	SCL2	I2C2_SCL	PB13
3	SDA2	I2C2_SDA	PB14
4	GND	GND	

H1 · SWD for ST-Link

used by 0.5.39

PZ254 2.54 mm right-angle, pin 1 = square pad · 2.54 mm

PIN	SIGNAL	NET	MCU
1	3V3	MCU_3V3	
2	SWDIO	SWDIO	PA13
3	SWCLK	SWCLK	PA14
4	GND	GND	

CN3 · Spare UART

not used by firmware

USART3 + I²C2 (unused) · JST GH 1.25 mm

PIN	SIGNAL	NET	MCU
1	5V	VOUT_5V	
2	TX3	USART3_TX	PB10
3	RX3	USART3_RX	PB11
4	SCL2	I2C2_SCL	PB13
5	SDA2	I2C2_SDA	PB14
6	GND	GND	

U138 · I²C3

not used by firmware

spare bus (unused) · JST GH 1.25 mm

PIN	SIGNAL	NET	MCU
1	5V	VOUT_5V	
2	SCL3	I2C3_SCL	PC0
3	SDA3	I2C3_SDA	PC1
4	GND	GND	

U140 · I²C3 header

not used by firmware

2.54 mm vertical, unused · 2.54 mm

PIN	SIGNAL	NET	MCU
1	3V3	MCU_3V3	
2	GND	GND	
3	SDA3	I2C3_SDA	PC1
4	SCL3	I2C3_SCL	PC0

S1–S4 · servo headers (U139, U143–U145)

not used by firmware

2.54 mm vertical, unused by firmware · 2.54 mm

PIN	SIGNAL	NET	MCU
1	S1 / S2 / S3 / S4	S1–S4	PA8 / PA9 / PA10 / PA11
2	NC	—	
3	GND	GND	

Bottom solder pads

SILK	NET	MCU	FUNCTION	REF
BAT1	BAT1	PC2	○ battery 1 voltage, divider ×21 or ×11	U132
BAT2	BAT2	PC4	○ battery 2 voltage, divider ×21 or ×11	U136
BZ-	BZ-	PA0	○ buzzer low side, S8050 switched by PA0	U121
CANH	CAN1_H		● CAN bus (same as U142-2)	U112
CANL	CAN1_L		● CAN bus (same as U142-3)	U113
C-RX	CAN1_RX	PB8	● MCU-side CAN logic, test point	U104

SILK	NET	MCU	FUNCTION	REF
C-TX	CAN1_TX	PB9	● MCU-side CAN logic, test point	U105
CUR1	CURR_ADC	PC3	○ current sensor 1, 0–3.3 V	U133
CUR2	CURR_ADC_2	PC5	○ current sensor 2, 0–3.3 V	U135
SCL2	I2C2_SCL	PB13	● sensor bus	U85
SDA2	I2C2_SDA	PB14	● sensor bus	U86
SCL3	I2C3_SCL	PC0	○ spare I²C	U124
SDA3	I2C3_SDA	PC1	○ spare I²C	U125
3V3	MCU_3V3		● 3.3 V out of the LDO	U110
CS	MS5611_CS	PA15	○ external MS5611 chip select	U137
S1	S1	PA8	○ servo / PWM 1 (TIM1_CH1)	U117
S2	S2	PA9	○ servo / PWM 2 (TIM1_CH2)	U118
S3	S3	PA10	○ servo / PWM 3 (TIM1_CH3)	U119
S4	S4	PA11	○ servo / PWM 4 (TIM1_CH4)	U120
MISO	SPI1_MISO	PA6	○ external MS5611 baro	U95
MOSI	SPI1_MOSI	PA7	○ external MS5611 baro	U94
SCK	SPI1_SCK	PA5	○ external MS5611 baro	U93
SWCLK	SWCLK	PA14	● SWD (same as H1-3)	U109
SWDIO	SWDIO	PA13	● SWD (same as H1-2)	U108
RX1	USART1_RX	PB7	● camera RX (same as CN1-3)	U90
TX1	USART1_TX	PB6	● camera TX (same as CN1-2)	U89
RX2	USART2_RX	PA3	● GNSS RX (same as CN2-3)	U101
TX2	USART2_TX	PA2	● GNSS TX (same as CN2-2)	U100
RX3	USART3_RX	PB11	○ spare UART	U84
TX3	USART3_TX	PB10	○ spare UART	U83
LED	W2815	PA1	○ WS2815 LED data (TIM2_CH2)	U122
5V	VOUT_5V		● 5 V rail (all connector pin 1s)	U99
5V	VOUT_5V		● 5 V rail (all connector pin 1s)	U126
5V	VOUT_5V		● 5 V rail (all connector pin 1s)	U130
BZ+	VOUT_5V		● buzzer high side = 5 V rail	U128
5V	VOUT_5V		● 5 V rail (all connector pin 1s)	U92
5V	VOUT_5V		● 5 V rail (all connector pin 1s)	U88
5V	VOUT_5V		● 5 V rail (all connector pin 1s)	U115
5V	VOUT_5V		● 5 V rail (all connector pin 1s)	U82
GND	GND		● ground	U102
GND	GND		● ground	U97
GND	GND		● ground	U129
GND	GND		● ground	U127
GND	GND		● ground	U131
GND	GND		● ground	U134
GND	GND		● ground	U91
GND	GND		● ground	U111
GND	GND		● ground	U87

SILK	NET	MCU	FUNCTION	REF
GND	GND		ground	U116

STM32L433RCT6 pin use

PORT	LQFP64	FUNCTION	GOES TO	0.5.39
PA2 / PA3	16 / 17	USART2 TX / RX	CN2 · GNSS 460800 (autobaud), filter build only	used
PB6 / PB7	58 / 59	USART1 TX / RX	CN1 · camera MAVLink tunnel 115200	used
PB8 / PB9	61 / 62	CAN1 RX / TX (AF9)	TJA1051 → U142 · DroneCAN node 42	used
PB13 / PB14	34 / 35	I²C2 SCL / SDA	MS4525 0x28 · MMC5983MA 0x30, both optional	used
PB1	27	GPIO out	LED1 status (active high)	used
PA13 / PA14	46 / 49	SWDIO / SWCLK	H1 programming	used
PH0 / PH1	5 / 6	HSE	8 MHz resonator → PLL 80 MHz	used
PB10 / PB11	29 / 30	USART3 TX / RX	CN3 · spare (schematic note: GPS)	spare
PC0 / PC1	8 / 9	I²C3 SCL / SDA	CN2 pins 4–5, U138, U140	spare
PC2 / PC3	10 / 11	ADC	BAT1 (×21 or ×11) / CURR1	spare
PC4 / PC5	24 / 25	ADC	BAT2 (×21 or ×11) / CURR2	spare
PA0	14	TIM2_CH1	buzzer via S8050 (BZ-)	spare
PA1	15	TIM2_CH2	WS2815 LED data	spare
PA5 / PA6 / PA7	21 / 22 / 23	SPI1 SCK / MISO / MOSI	bottom pads, MS5611	spare
PA15	50	GPIO	MS5611 chip select pad	spare
PA8 – PA11	41 – 44	TIM1_CH1 – CH4	S1 – S4 headers	spare
PH3-BOOT0	60	BOOT0	10 kΩ to GND: boots from flash	used

Generated by `tools/render_l433_pinout.py` from the EasyEDA project `APMPX4 CAN-JSTGH.eprj2` (PCB1 placement and netlist), for the firmware builds `apmpx4_can_l433rc_dronecan` and `apmpx4_can_l433rc_dronecan_nognss` 0.5.39.